



Indian Institute of Information Technology, Pune

LLM-Based Clinical Intake System

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INTRODUCTION

1. Manual intake and scattered data slow down diagnosis and care delivery.
2. AI-driven interaction captures patient symptoms quickly and efficiently.
3. Automated SOAP notes give doctors concise, actionable insights.
4. The platform integrates emergency support, medication safety checks, and report analysis to deliver a comprehensive, end-to-end healthcare solution.



MOTIVATION

01

Patients struggle to clearly communicate symptoms, due to standard checkboxes and scales

02

Doctors have multiple patients, delaying possible quicker diagnoses

03

Emergency responders do not have access to critical patient details

04

Lack of a standardized doctor-patient platform, with fragmented report analysis and lack of medication safety

LITERATURE REVIEW

Author & Year	Title	Method/Model	Learning	Key Contributions	Limitations
Laurent Boyer et. al, 2025 [1]	Reimagining patient reported outcomes in the age of generative AI	Proposes integration of Large Language Models (LLMs) with traditional psychometric models like IRT and CAT for AI-driven PRO assessment.	Uses Generative AI (LLMs) based on deep learning and NLP for conversational and narrative understanding.	Introduces a shift from static questionnaire based PROs to adaptive, narrative based AI systems for personalized patient symptom extraction and analysis.	Lacks established validation frameworks, psychometric reliability proof, and safeguards against bias and hallucinations in clinical deployment.
Mukul Bedwa et. al, 2025 [2]	Optimizing RAG-based LLMs for healthcare question answering tasks	Applies Large Language Models (LLMs) for structured and unstructured healthcare data analysis.	Uses Transformer based deep learning with pretraining and fine tuning on medical datasets.	Demonstrates how LLMs can extract, summarize, and interpret clinical information to support automated medical decision systems.	Highlights concerns regarding data privacy, model bias, hallucinations, and limited clinical validation.
Tsai-Ling Liu et. al, 2026 [3]	AI Powered Clinical Documentation and Clinicians' Electronic Health Record Experience	Evaluated an AI powered clinical documentation system integrated with Electronic Health Records (EHRs) in a real clinical setting.	Uses NLP based AI models (likely LLM assisted documentation support) for automated clinical note generation.	Demonstrates that AI assisted documentation can reduce clinician burden and improve EHR experience, supporting real world clinical AI deployment.	Nonrandomized design with potential selection bias and limited generalizability across diverse healthcare settings.

LITERATURE REVIEW

Author & Year	Title	Method/Model	Learning	Key Contributions	Limitations
Isabella Joy De Vere Hunt et. al, 2025 [4]	A framework for considering the use of generative AI for health	Provides a conceptual, expert-informed framework outlining principles and risks for evaluating generative AI (mainly LLMs) applications in healthcare.	Framework built from expert interviews and roundtable deliberations rather than empirical model training.	Highlights principles for safe, equitable, and impactful integration of AI in health, including aligning use cases to AI strengths and robust evaluation practices.	Lacks empirical tests or quantitative validation of the proposed framework in real healthcare deployments.
Elias Hussain et. al, 2023 [5]	Natural Language Processing in Electronic Health Records in relation to healthcare decision-making: A systematic review	Comprehensively reviews ML, DL, and traditional NLP techniques applied to unstructured EHR text for tasks like classification, entity recognition, and information extraction.	Covers studies using a range of machine learning and deep learning approaches (e.g., SVM, LSTM, CNN) for clinical text analysis in EHRs	Highlights how NLP methods can transform unstructured clinical notes into structured insights to support healthcare decision-making workflows.	Identifies major challenges including lack of annotated data, inadequate model evaluation, and data imbalance issues across existing NLP-EHR studies.
Xuexing Luo et. al, 2025 [6]	AI in Medical Questionnaires: Scoping Review	Surveys existing literature on AI-enabled questionnaire methods, including rule-based algorithms, classical NLP, and early generative models for patient-reported data capture.	Focuses on machine learning and natural language processing approaches used to interpret, adapt, and automate responses in medical questionnaires.	Shows how AI can improve the accuracy, efficiency, and personalization of medical questionnaires, enabling better data capture for clinical decision support.	Highlights gaps like limited clinical validation, heterogeneity of methods, and lack of standardized evaluation metrics across studies.

RESEARCH GAPS

- No reliable safeguards to prevent harmful AI errors in medical outputs.
- Lack of tools to generate concise SOAP summaries for clinical assistance.
- No unified system linking patient check-ins with long-term health tracking.
- No automated system to ensure treatment adherence and continuity of care.
- AI outputs are not validated against trusted medical databases.
- No secure way to share critical patient data with first responders.
- Clinical triage is not integrated with real-time hospital discovery and routing.



OBJECTIVES

01

To develop an AI-driven system that automates patient intake and accurately captures symptoms through natural interaction

02

To generate structured clinical summaries (SOAP notes) that assist doctors in faster and more informed decision-making.

03

To provide emergency support via QR-based medical profiles and real-time nearby hospital detection.

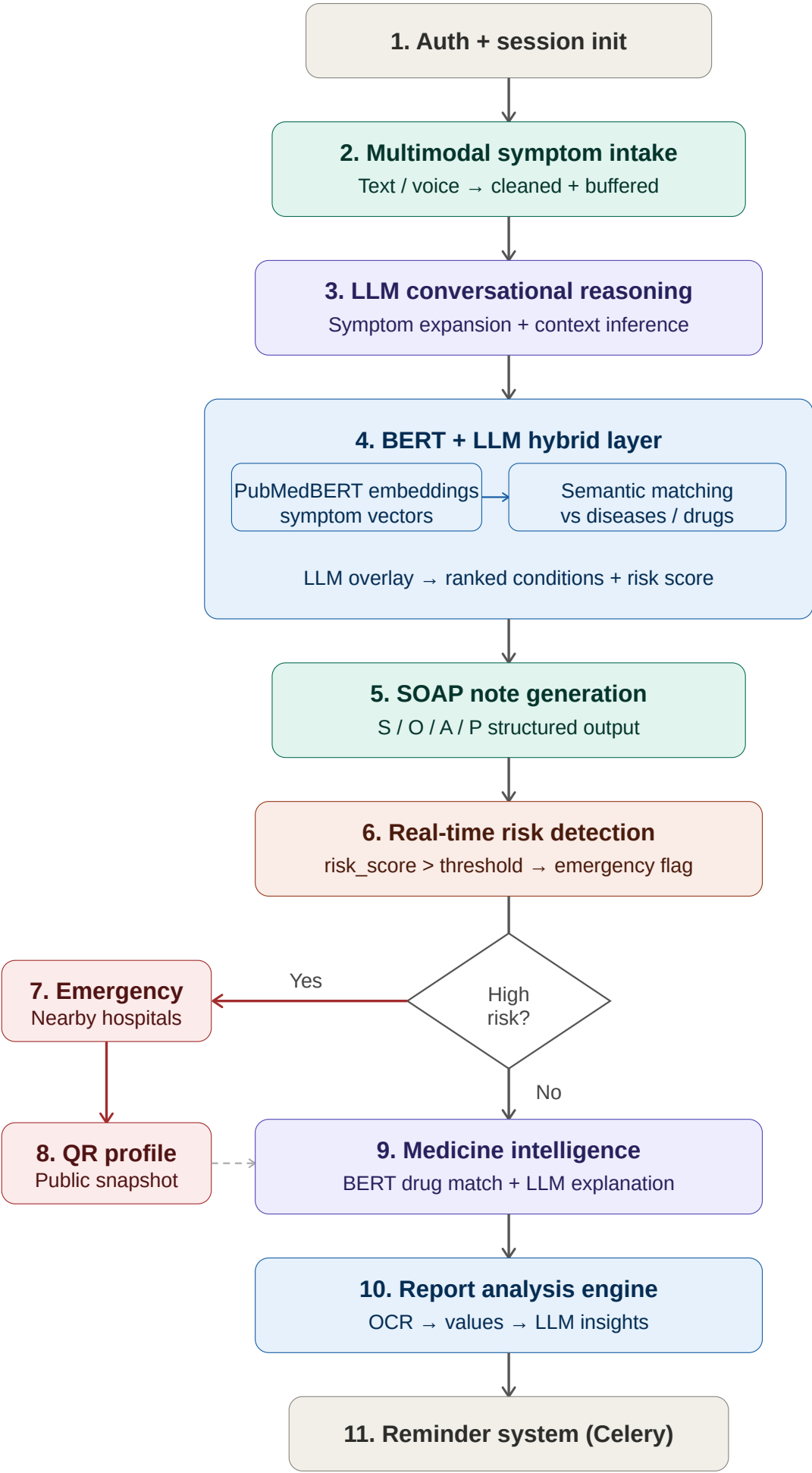
04

To enhance medication safety by analyzing drug information, side effects, and potential interactions using AI.

METHODOLOGY

PATIENT ENTRY FLOW AND AI PATIENT INTAKE SYSTEM

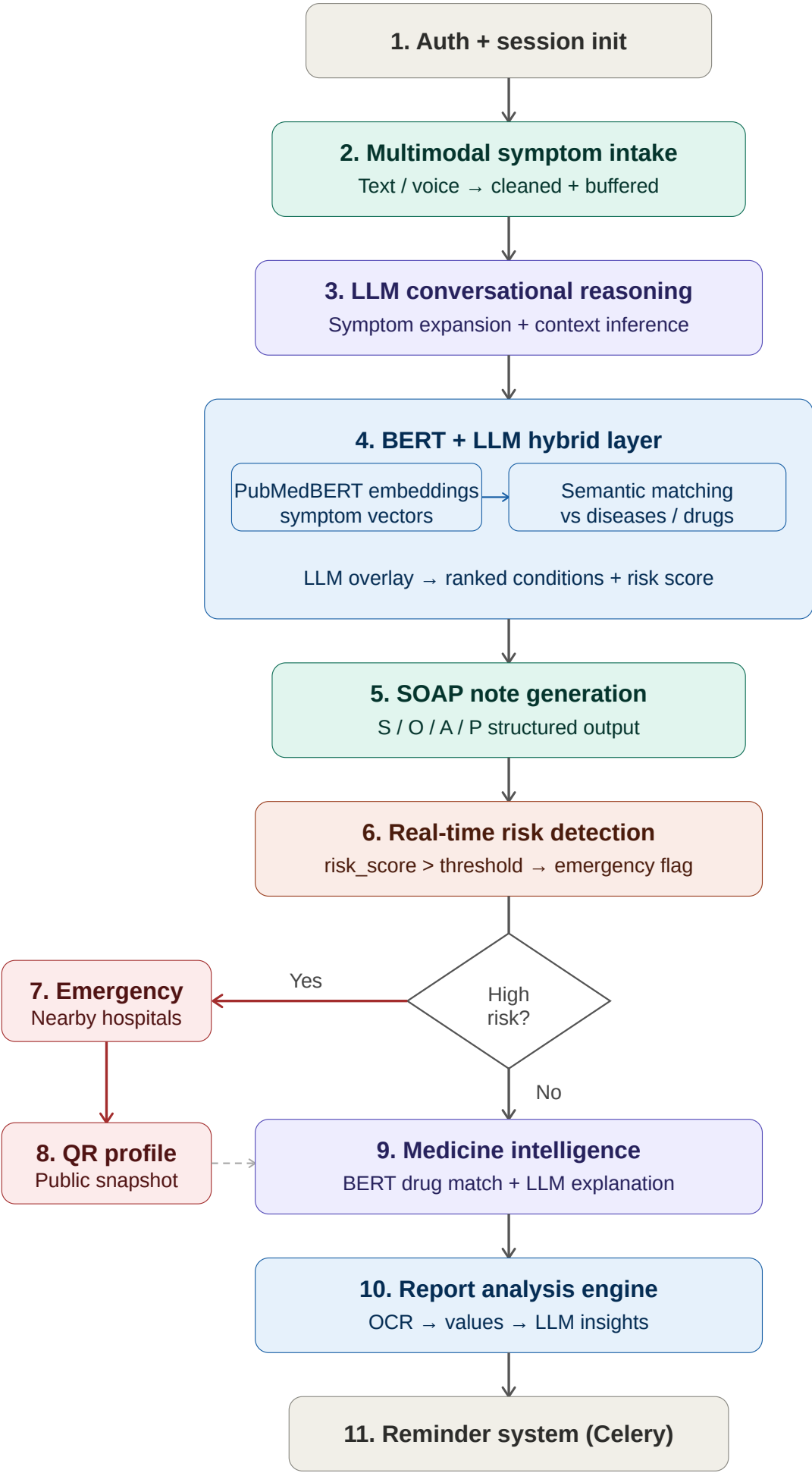
- **User Authentication:** Secure login/signup using JWT-based authentication.
- **Profile Creation:** Basic details: age, gender, blood group
- **Medical history:** allergies, chronic conditions, medications
- **Multimodal Symptom Intake:** Chat-based conversational interface and Voice-to-text input (Web Speech API)
- **Dynamic Questionnaire:** AI adapts follow-up questions based on responses
- **Output:** Structured symptom dataset for downstream AI processing



METHODOLOGY

AI PROCESSING LAYER & EMERGENCY HANDLING

- **LLM-Based Analysis (Groq/Gemini):** Interprets symptoms and medical history.
- **Hybrid Intelligence Layer:** PubMedBERT embeddings + semantic matching + LLM reasoning for ranking conditions and interactions.
- **SOAP Note Generation:** Structured output: Subjective, Objective, Assessment, Plan.
- **Real-Time Risk Detection:** Assigns risk levels (low / medium / high).
- **Emergency Protocol (High Risk):** Triggers alerts, suggests nearby hospitals (OpenStreetMap API), and activates emergency mode.
- **QR Emergency Profile:** Shareable snapshot with critical data (allergies, medications, blood group).



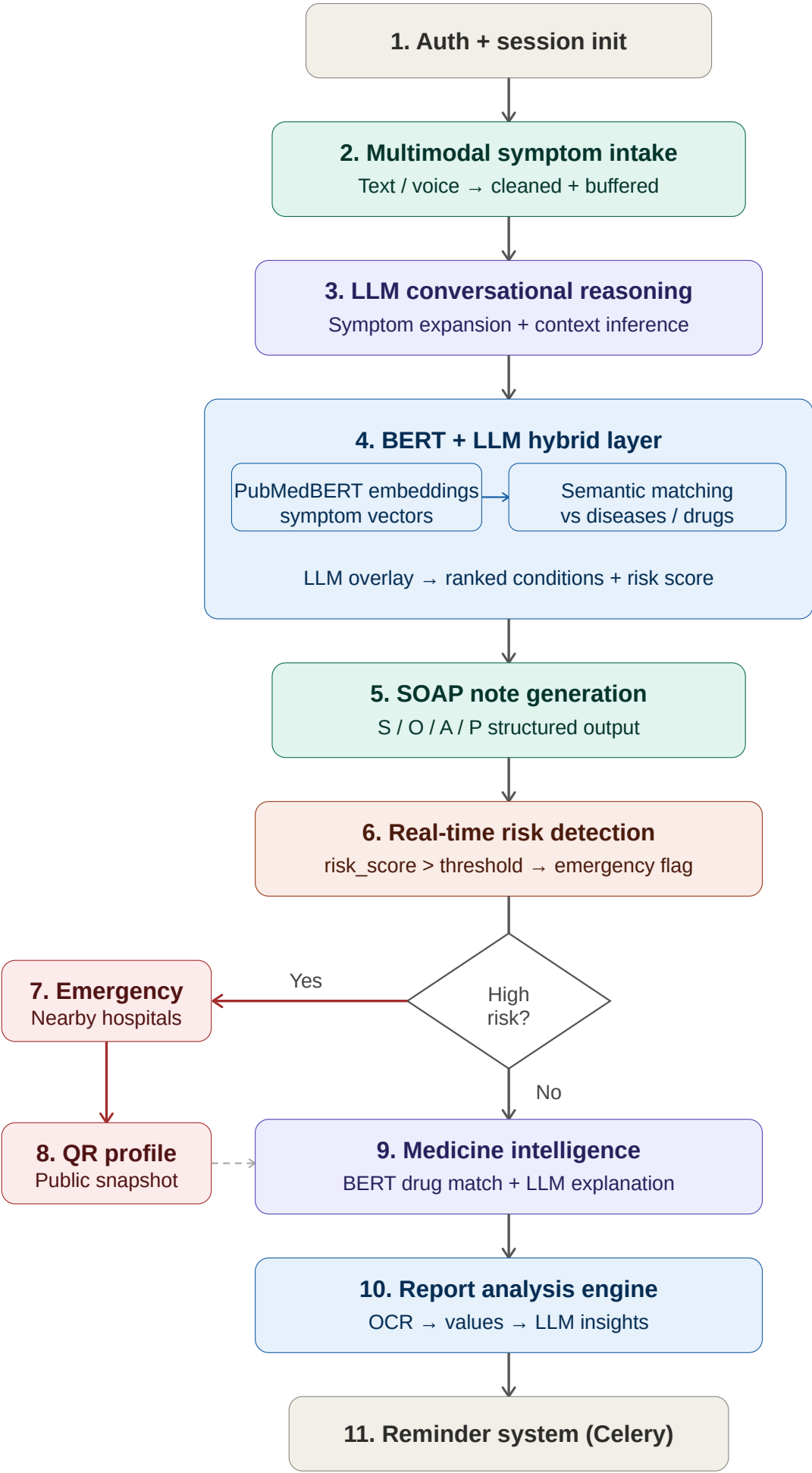
METHODOLOGY

MEDICINE INTELLIGENCE & SAFETY LAYER

- **AI Medicine Assistant:** Uses NLP + PubMedBERT to understand drugs and patient context.
- **Medicine Info Box:** Context-aware UI shown during symptom analysis, prescriptions, and search.
- **Displays:** benefits, side effects, interactions, contraindications, and usage guidance.
- **Semantic Risk Check:** Matches symptoms with side effects and checks new drugs against existing medications.
- **AI Alerts:** Flags adverse reactions, allergy conflicts, and unsafe combinations.

MEDICAL REPORT ANALYSIS ENGINE

- **Report Upload:** Supports PDFs and images
- **OCR + AI Parsing:** Extracts lab values and key indicators
- **Insight Generation:** Converts raw data → simplified explanations
- **Trend Tracking:** Compares historical reports over time



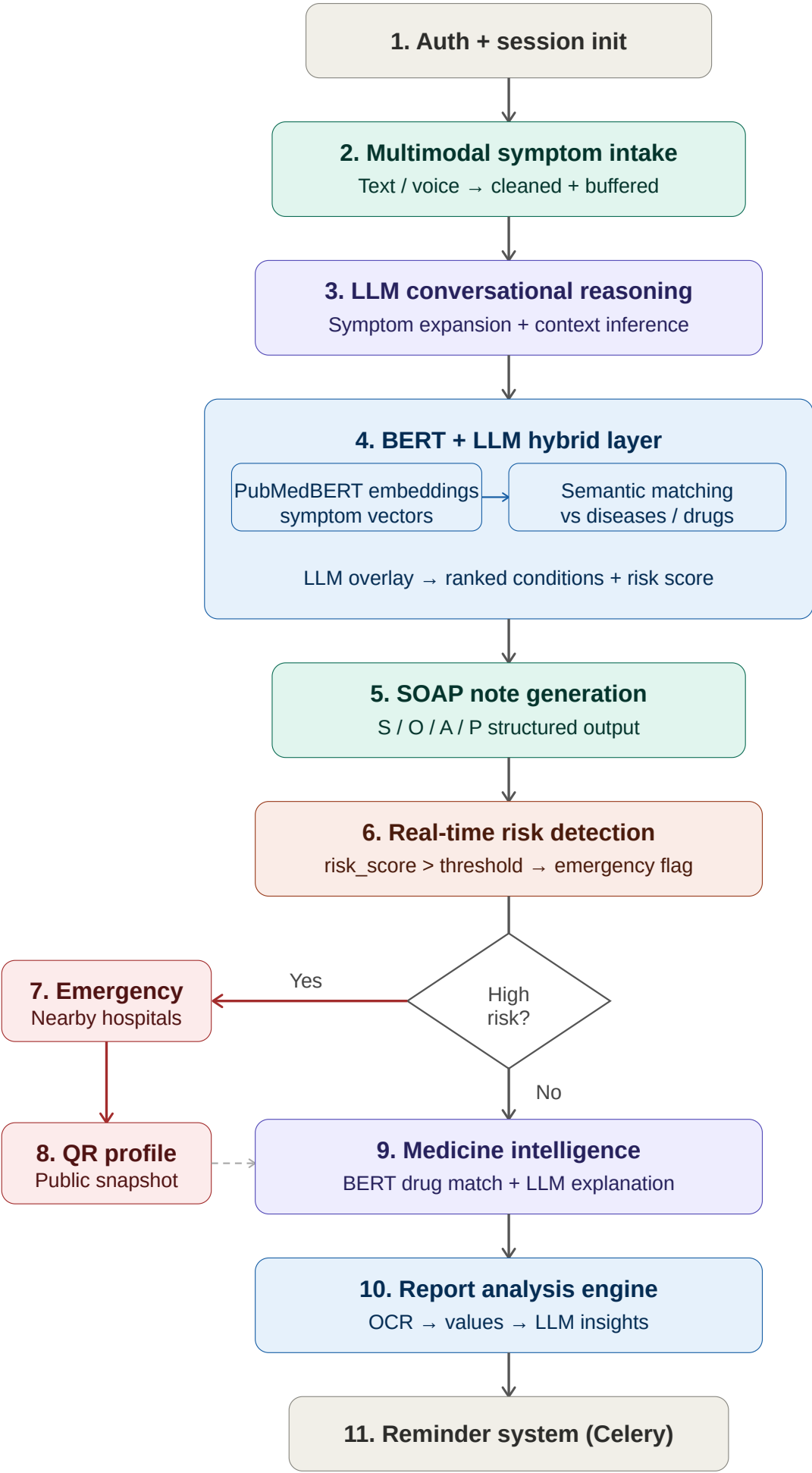
METHODOLOGY

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PATIENT DASHBOARD

- **Health Visualization:** Charts (Recharts) for vitals & lab metrics
- **Medication Management:**
 - Reminders (scheduled + urgent alerts)
 - Integrated Medicine Info Box for each drug
- **Appointments Tracking:** Upcoming consultations & follow-ups



METHODOLOGY

DOCTOR LOGIN & DASHBOARD

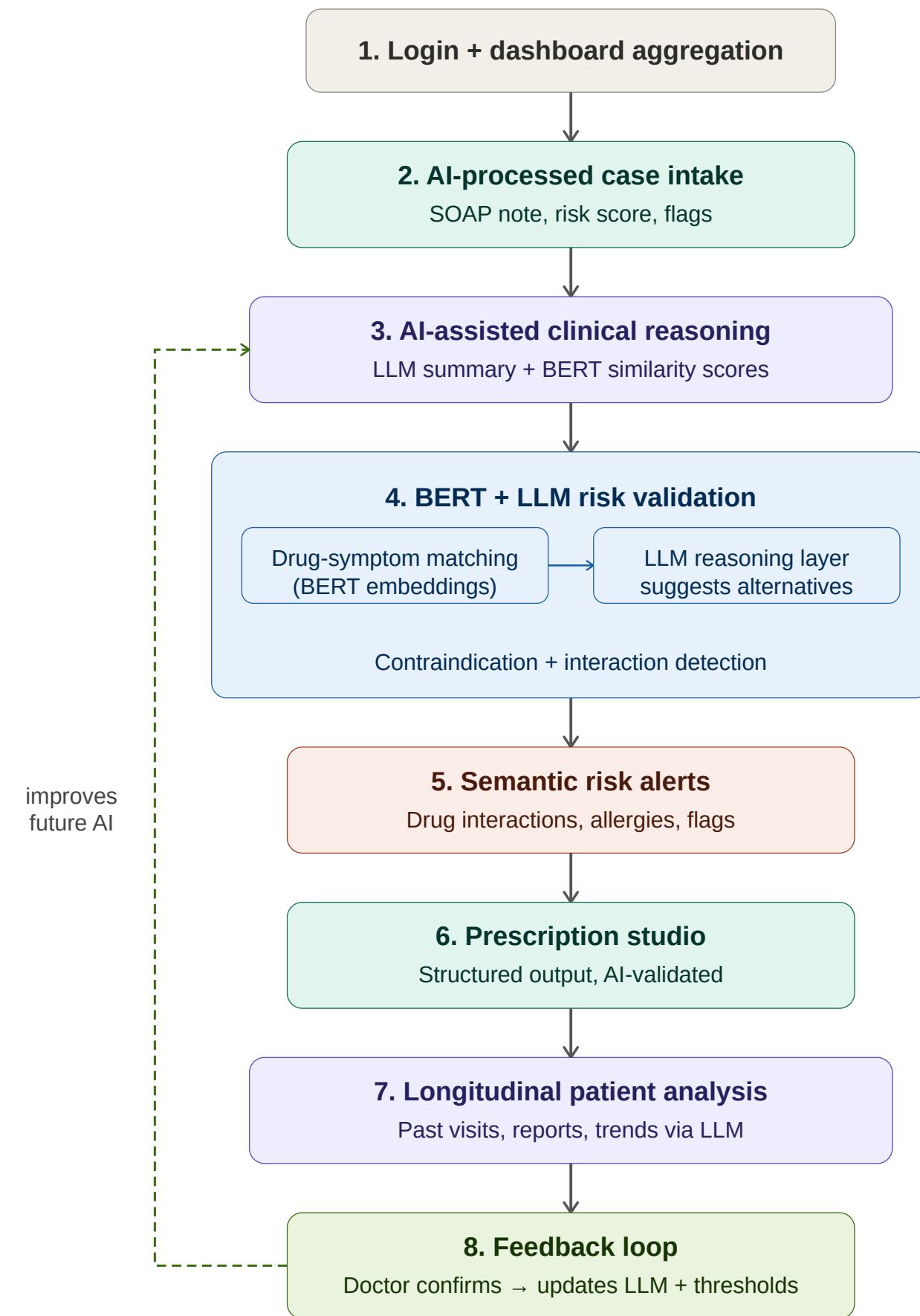
- Secure authentication
- Access to patient queue
- Real-time case updates

AI CLINICAL SUMMARY

- Auto-generated SOAP notes from patient input
- Saves consultation time
- Structured clinical view:
 - Subjective
 - Objective
 - Assessment
 - Plan

DECISION SUPPORT SYSTEM

- Semantic Risk Check (PubMedBERT)
- Flags:
 - Drug side effects vs symptoms
 - Potential adverse reactions



METHODOLOGY

PRESCRIPTION STUDIO

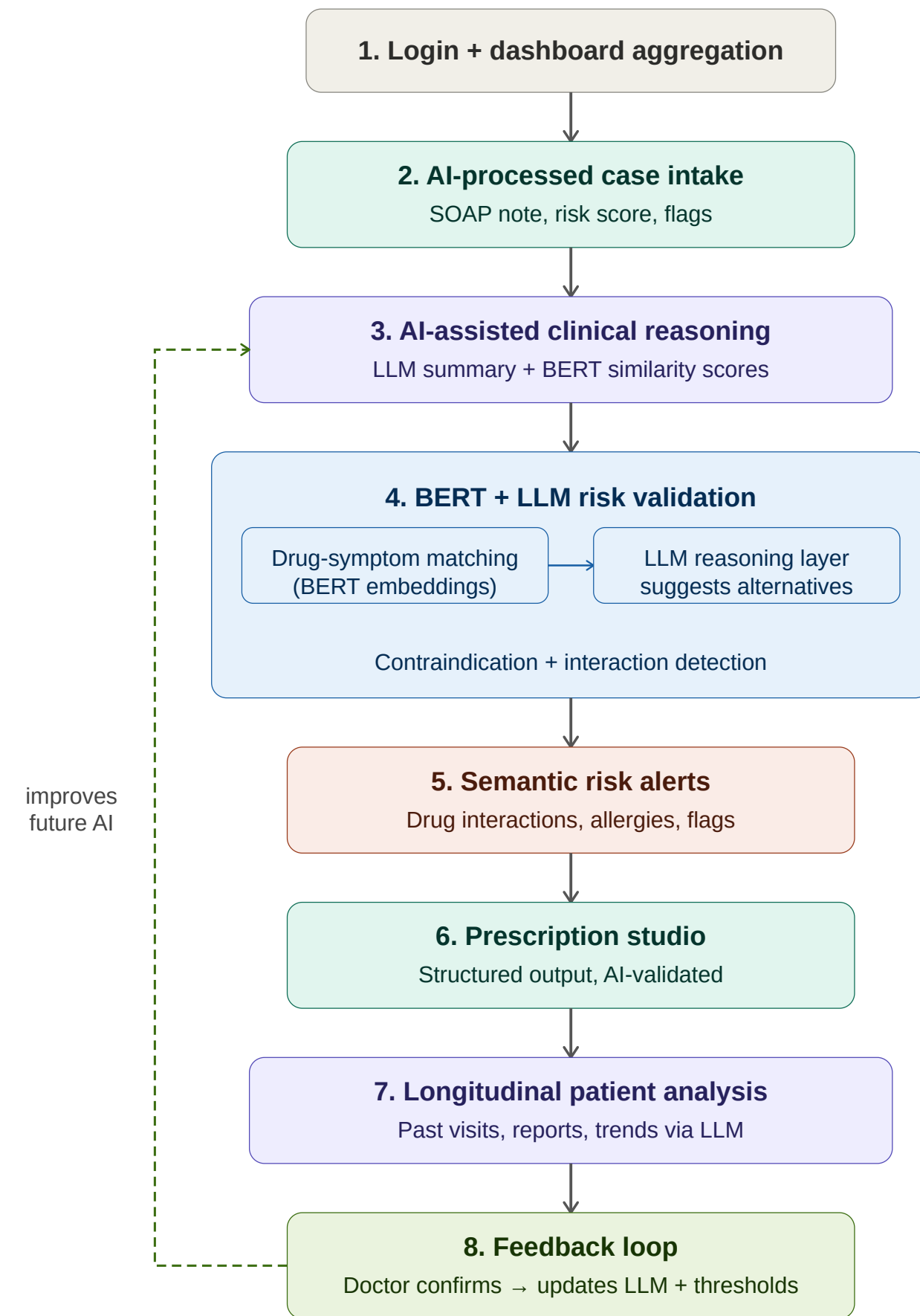
- Digital prescription generation
- Features:
 - Dosage control
 - Frequency selection
 - Custom instructions

PATIENT MONITORING

- Access patient history and view:
 - Uploaded reports
 - Trends in vitals
 - Continuous care support

EMERGENCY ACCESS VIA QR

- Scan patient QR code and get instant:
 - Allergies
 - Blood group
 - Medications



■ EXPECTED OUTCOMES

**Structured Patient
Summary**

System converts
free-text patient
narrative into clear
structured data

**Medical History
Snapshot**

Uploaded reports are
summarized into concise
historical medical
information for quick
review.

**Visit Change
Detection Report**

Automatic comparison
between current and
previous visits showing
improvements or worsening
conditions.

**Doctor-Ready
Clinical View**

Single dashboard view
displaying Current State,
History, and Changes –
reducing consultation time
and cognitive load.

■ FUTURE SCOPE

Clinical Decision Support: "Similar Cases"

AI uses vector embeddings to analyze similar patient cases and suggest commonly recommended tests or next diagnostic steps.

It acts as a smart second opinion, helping doctors reduce oversight and make faster, data-driven clinical decisions.

Automated Imaging Insights

AI processes complex medical imaging like X-rays and sonography to instantly detect subtle anomalies and highlight critical areas of concern. This intelligent screening provides doctors with rapid, data-driven insights to confidently accelerate their diagnostic workflows.

Automated Clinical Scribing Description:

AI securely transcribes live audio conversations between doctors and patients to automatically generate accurate, structured clinical notes. These summaries are instantly saved to the patient's visit history, eliminating manual documentation and allowing physicians to focus entirely on patient care.

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THANK YOU